

UNIT 1:

1. Write a note on various layers of the blockchain with a neat diagram.
2. Differentiate between centralised and decentralized systems.
3. Define blockchain and list its various advantages and disadvantages with explanation.
4. Illustrate various applications of blockchain in various sectors.
5. Explain in detail about various fields using which blockchain technology is developed with a suitable diagram.
6. Write a detailed note on the importance of blockchain.
7. Discuss evolution and history of blockchain.

UNIT 2:

1. Explain SHA-256 cryptographic hash function in detail with suitable diagram.
2. What is a hash function? List and explain various properties related to hash function.
3. Differentiate between asymmetric and symmetric key cryptographic systems.
4. Write a note on the RSA algorithm.
5. Explain various cryptographic system objectives in detail.
6. Write a note on digital signature algorithm with suitable diagram.
7. Explain about the Elliptic Curve Cryptography in detail with suitable diagram.
8. Explain about the Diffie Hellman Key Exchange with a suitable example.
9. What is symmetric key cryptography? Explain in detail about DES cryptography.
10. Explain in detail about the AES algorithm.
11. What is MAC and HMAC? Explain in detail some widely used important MAC strategies.

UNIT 3:

1. What is consensus? Explain in detail about the Byzantine Generals Problem.
2. Differentiate between CFT and BFT consensus algorithms.
3. Explain the working of CFT algorithm with a suitable example.
4. What factors should be considered while choosing an appropriate consensus algorithm for a blockchain application?
5. Discuss the principles and working of BFT algorithms.
6. Describe the design considerations and goals of a consensus algorithm.
7. Classify different types of consensus algorithms.

UNIT 4:

1. Write in detail about the block structure in the Bitcoin blockchain with a suitable diagram.
2. What is the difficulty target in Bitcoin blockchain? Explain its working in detail.
3. Explain in detail various fields in Bitcoin blockchain blocks with a neat diagram.
4. What is block mining? Explain the process of block mining in Bitcoin in detail.
5. Explain in detail about the Merkle Tree data structure in Bitcoin blockchain.
6. Explain Bitcoin wallets with the help of a suitable diagram.
7. Explain in detail about the Bitcoin scripts with suitable diagrams.
8. Explain about the Bitcoin transactions in detail.
9. Explain about the Bitcoin network with a neat diagram.
10. Write a brief history of money and the dawn of Bitcoin.
11. Discuss the network discovery for a new node in Bitcoin.
12. What is a Genesis Block in Bitcoin? Explain in detail.

UNIT 5:

1. Differentiate between Bitcoin and Ethereum.
2. Discuss EVM in detail with its neat architecture diagram.
3. Discuss block structure of Ethereum Blockchain with a neat diagram.
4. Explain in detail about the transactions and message structure in Ethereum.
5. Explain: 1) Merkle Patricia Tree 2) RLP Encoding.
6. What are smart contracts in Ethereum? Explain in detail.
7. Explain how Ethereum forms as a Next-Gen blockchain.
8. What are Ethereum Accounts? Explain about them in detail.
9. Discuss in detail about the design philosophy of Ethereum.
10. Discuss about the Trie Usage in Ethereum.

UNIT 6:

1. What is Solidity? Explain in detail. Create a smart contract to calculate the factorial of a given number.
2. What are decentralized applications? Explain in detail with suitable examples.
3. Explain the process of interacting with the Bitcoin blockchain with the help of a suitable diagram.
4. Discuss the key steps involved in blockchain application development.
5. Write a detailed note on sending transactions in Ethereum.
6. Explain how Ethereum allows programmatic interaction.